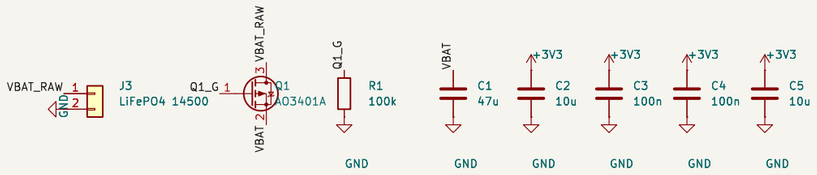
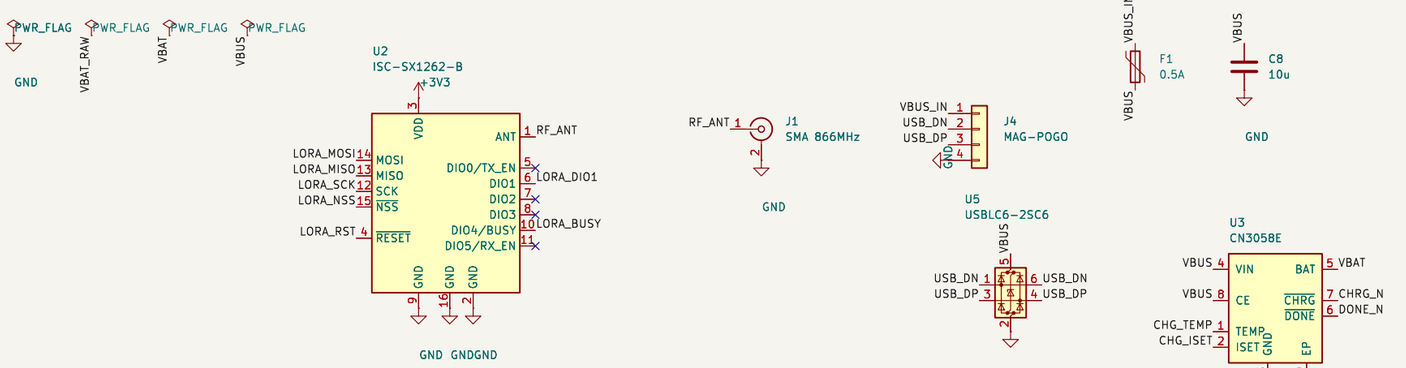
Power: LiFePO4 14500 -> Q1 reverse protection -> VBAT -> U4 RT9080 LDO (2 uA Iq) -> +3V3. VBAT reaches 3.6 V while charging.

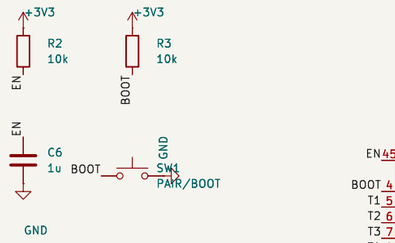


LoRa: ISC-SX1262-B in Ra-01 footprint - same module and pin map as WPC (BUSY = pad 10). SMA -> pigtail -> antenna on a 1.5 m pole above the canopy.

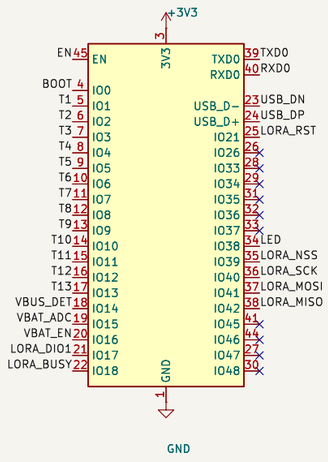
v0.2 field charging: J4 magnetic socket -> F1 -> U3 CN3058E (0.31 A, 3.6 V).

U5 = USB ESD. RED = charging, GREEN = full. VBUS_DET on IO14 wakes the S3.

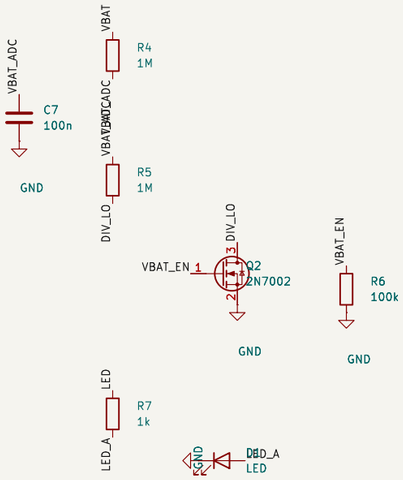


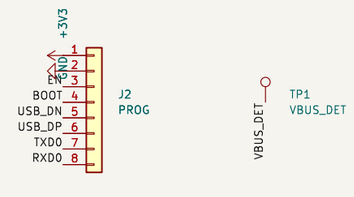


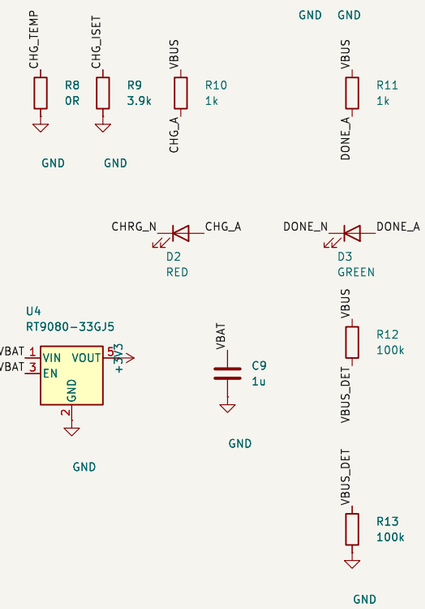
U1
ESP32-S3-MINI-1-N8



Battery sense: Q2 enables the 1M/1M divider (from VBAT) only while sampling VBAT_ADC (ADC2, WiFi off).

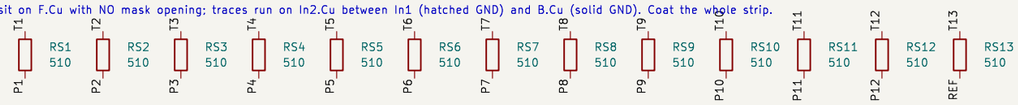






Sensing strip: E1..E12 = 8x18 mm pads on 20 mm pitch (P1 = +7..+5 cm ... P12 = -15..-17 cm), E13 = dry reference, E14 = GND water-return pad.

Pads sit on F.Cu with NO mask opening; traces run on In2.Cu between In1 (hatched GND) and B.Cu (solid GND). Coat the whole strip.



RS1..RS13 = 510R touch series resistors, placed at the strip neck in the head.



Generated by tools/gen_sch.py - edit design.py and regenerate, or take over by hand

Agri Sensors and Controls

Sheet: /

File: AWD-FieldNode.kicad_sch

Title: AWD field node - segmented capacitive level strip

Size: A3	Date: 2026-10-06	**Rev: 0.1**
KiCad E.D.A. 9.0.7		Id: 1/1